Week 14 — Extended Reading: Offline RL

STUDY-RL

May 21, 2026

This companion reads alongside `notes.pdf`. Each entry: full citation, one-paragraph summary, "Why it matters for this week", and an annotated key passage.

1 Levine, Kumar, Tucker, Fu (2020) — Offline RL Tutorial

S. Levine, A. Kumar, G. Tucker, J. Fu. *Offline Reinforcement Learning: Tutorial, Review, and Perspectives on Open Problems.* arXiv:2005.01643, 2020.
Summary. The canonical reference for the field. Formalises the problem; surveys distribution shift, importance sampling, BCQ/BEAR/CQL/AWR; proposes a taxonomy of methods by where they place the constraint (policy vs. critic vs. density). Long but dense; chapters 4 and 6 are mandatory.
Why it matters for this week. It is the canonical mental model: all subsequent papers are positioned with respect to this taxonomy.

> "The fundamental challenge in offline RL is one of distributional shift: while we optimise the policy under one distribution (the dataset $\mathcal{D}$), the policy will visit a different distribution at evaluation time."

Commentary. This one sentence is the entire field. Every algorithm in the literature is a different answer to: *which divergence between d^π and d^{π_β} are we bounding, and how?*

2 Kumar et al. (2020) — Conservative Q-Learning

A. Kumar, A. Zhou, G. Tucker, S. Levine. *Conservative Q-Learning for Offline RL.* NeurIPS 2020, arXiv:2006.04779.
Summary. Introduce the now-standard "learned Q should be a lower bound on V^π" criterion. The CQL penalty $\log \sum_a \exp Q(s,a) - \mathbb{E}_{\pi_\beta} Q(s,a)$ added to Bellman regression yields a lower-bound estimator under mild assumptions on the data coverage. Strong on D4RL-MuJoCo.
Why it matters. It moved the field from "policy constraint" to "value constraint". Cleaner objective, fewer moving parts than BCQ/BEAR.

> "By learning a conservative value function, CQL prevents the policy from exploiting overestimation errors during the policy improvement step."

Commentary. "Prevents" is overselling: it dampens. Tuning α is where production bites; in practice we use the Lagrangian variant.

3 Kostrikov et al. (2021) — Implicit Q-Learning

I. Kostrikov, A. Nair, S. Levine. *Offline RL with Implicit Q-Learning.* arXiv:2110.06169, 2021.
Summary. Replace explicit Q-on-out-of-support sampling with an *expectile* regression on the value: V regresses to the upper expectile of $Q(s, a \sim \pi_\beta)$, so it estimates "best in-support" without ever naming the out-of-support action. Pairs with AWR/AWAC for policy extraction. Faster, simpler, competitive with CQL.
Why it matters. IQL is the current default in many real-world deployments precisely because there are no OOD samples needed at training time — the action space can be arbitrary.

> "Our key insight is that we can perform the policy improvement step implicitly by treating the value function as the upper expectile of the action-value function under the data distribution."

Commentary. The word "implicitly" carries the whole idea: the policy is never used to query Q at unseen actions. Compare to AWAC, which still samples actions from π.

4 Tarasov et al. (2023) — ReBRAC

D. Tarasov, V. Kurenkov, A. Nikulin, S. Kolesnikov. *Revisiting the Minimalist Approach to Offline RL.* arXiv:2305.09836.
Summary. A careful re-engineering of TD3+BC: higher discount, layer-norm in critics, per-step BC weight tuned to $\mathbb{E}[Q]$. No new ideas, just correct execution. Sets the strongest non-transformer scores on D4RL as of late 2023.
Why it matters. Lesson: many offline RL "algorithmic" gains were hyperparameter gains. Useful sanity check before reaching for the heavy tools.

> "Many of the recent advances in offline RL can be attributed to a few well-chosen implementation details rather than to the algorithmic novelty."

Commentary. A polite rebuke of the field. Cite this paper whenever a new SOTA claim arrives without ablations.

5 Voloshin et al. (2021) — OPE Empirical Study

C. Voloshin, H. Le, N. Jiang, Y. Yue. *Empirical Study of Off-Policy Policy Evaluation.* NeurIPS Datasets and Benchmarks, 2021, arXiv:1911.06854.
Summary. A systematic comparison of IS, weighted-IS, doubly robust, FQE, MAGIC, DICE estimators across a portfolio of MDPs. Verdict: **FQE** is the single most reliable estimator when a function approximator is available; SNIPS is the best "no model" baseline; doubly-robust requires careful nuisance tuning.
Why it matters. OPE is what stands between an offline-trained policy and a production deployment. This paper is the "which estimator should I use" cheat-sheet.

> "No single OPE method dominates across all settings; the choice depends on horizon length, state-space dimensionality, and how well the behaviour policy is known."

Commentary. The horizon-length collapse is the key practical point: above roughly 50 steps, IS-family estimators are useless; FQE or DICE is mandatory.

Sourced from `research/offline_modelbased_sota.md` (track A, sections A.1–A.5, A.15–A.16).